

LYCEUM OF THE PHILIPPINES UNIVERSITY – LAGUNA

College of Business and Accountancy

MIDTERM ASSESSMENT — TEST II PRACTICAL

Bayanihan Bus Lines · 45 points · 120 minutes

One task per sheet. Write your EVIDENCE and DECISION here and hand it in.

Name: Sanchez, Eliah Therrien Estrella Drawn at: 10:18 AM Finished: 10:28 AM

BAND A · TASK A1 · 8 POINTS

A single operation with one correct outcome

What did you just load?

1. Load BayanihanBus_Tickets.xlsx.
2. Report the number of rows and the number of columns.
3. Count the DISTINCT values of TicketID and compare it with the row count.

EVIDENCE

Rows, columns, distinct TicketID, and the difference between the last two.

Columns: 7

Rows: 18,329

Distinct TicketID: 18,298

The difference: 31

DECISION

Written, in full sentences.

The Tickets table contains 18,329 rows and 7 columns. It has 18,298 distinct TicketID values, which is 31 fewer than the total number of rows. Therefore, TicketID is not unique across all rows because some TicketID values are duplicated.

Applied steps renamed? NO Query name: Tickets

Name: Sanchez, Eliah Therrien Estrella Drawn at: 10:29 AM Finished: 10:40 AM

BAND A · TASK A2 · 8 POINTS

A single operation with one correct outcome

The column that cannot be used

4. Load BayanihanBus_Trips.xlsx.
5. Report the number of rows.
6. Report the data type Power Query assigns to OdometerKM.
7. State in one sentence why that type is a problem.

EVIDENCE

Row count, the assigned type, and the consequence.

Rows: 1,200

OdometerKM data type before Changed Type: Any

Consequence: There are values that would contain commas.

DECISION

Written, in full sentences.

The OdometerKM column data type should be changed into a numeric type, specifically a whole number (Int64)

Applied steps renamed? NO Query name: Trips

Name: Sanchez, Eliah Therrien Estrella Drawn at: 10:40 AM Finished: 10:54 AM

BAND A · TASK A3 · 8 POINTS

A single operation with one correct outcome

How many tickets to a trip?

8. Load both Tickets and Trips.
9. Group the tickets by TripID and count the rows in each group.
10. Report the minimum, the maximum and the mean.

EVIDENCE

Min, max and mean tickets per trip — and one sentence on what this proves.

Min: 6

Max: 26

Mean: 15.261

It proves that the average tickets per trip is 15 tickets per trip

DECISION

Written, in full sentences.

The number of tickets ranges from 6 to 26, with an average of 15.261 tickets per trip. This shows that the trips have different numbers of tickets.

Applied steps renamed? NO Query name: Tickets

BAND B · TASK B1 · 10 POINTS

Two steps where the order matters

Make the odometer usable

11. OdometerKM will not convert to a whole number. Repair it.
12. Report how many values contained a comma.
13. Report the minimum and maximum odometer reading after the repair.

EVIDENCE

The comma count, the min and max, and the order of your two steps.

OdometerKM converted to a whole number for me.

Values contained a comma: 19

Minimum odometer: 48,000

Maximum odometer: 611,741

Order: Calculated Minimum -> Calculated Maximum

DECISION

Written, in full sentences.

The OdometerKM column should be cleaned and converted to a numeric type because some values contain commas and may not be treated correctly as numbers.

Applied steps renamed? NO Query name: Trips

Name: Sanchez, Eliah Therrien Estrella Drawn at: 11:07 AM Finished: 11:18 AM

BAND B · TASK B2 · 10 POINTS

Two steps where the order matters

More routes than exist

14. RouteID in Trips shows more routes than the Routes table contains.
15. Repair the formatting variants — delete nothing.
16. Report the distinct count before and after, and how many rows you repaired.

EVIDENCE

Before, after, and rows touched.

Before rows touched: 16

After rows touched: 12

Rows repaired: 21

DECISION

Written, in full sentences.

The RouteID formatting variants were repaired without deleting any rows. The distinct RouteID count decreased from 16 to 12, matching the number of valid routes in the Routes table.

Applied steps renamed? NO Query name: Trips

BAND B · TASK B3 · 10 POINTS

Two steps where the order matters

What the duplicates were worth

17. Remove duplicate rows from Tickets.
18. Report the row count before and after.
19. Report total FareAmount before and after, and the difference.

EVIDENCE

Both row counts, both revenue figures, and the peso difference.

Row count before: 18, 329

Row count after: 18,298

FareAmount before: PHP 2,528,647.94

FareAmount after: PHP 2,523,700.41

Difference: PHP 4,947.53

DECISION

Written, in full sentences.

Removing duplicate TicketID rows reduced the table from 18,329 to 18,298 rows. The total FareAmount decreased by ₱4,947.53, showing that the duplicate rows were contributing to the total revenue.

Applied steps renamed? NO Query name: Tickets

Name: Sanchez, Eliah Therrien Estrella Drawn at: 11:23 AM Finished: 11:36 AM

BAND C · TASK C1 · 12 POINTS

A diagnostic — find it, count it, say what it costs

A bus that is not in the fleet

20. Using a left anti join, find every trip whose BusID is not in the Buses table.
21. Count them and name the value.
22. State what happens to those trips in a report grouped by bus type.

EVIDENCE

Count, value, and the silent consequence.

Count: 17

BusID not in the Buses table: BU099

What happens: The 17 trips with no matching BusID in the Buses table cannot be assigned a BusType.

DECISION

Written, in full sentences.

There are 17 trips using BusID BU099, but BU099 does not exist in the Buses table. These trips should not be deleted but the missing bus information should be identified so that they could be assigned to a BusType for a report.

Applied steps renamed? NO Query name: Trips (2)

Name: Sanchez, Eliah Therrien Estrella Drawn at: 11:36 AM Finished: 11:47 AM

BAND C · TASK C2 · 12 POINTS

A diagnostic — find it, count it, say what it costs

A ticket for a trip that never ran

23. Using a left anti join, find every ticket whose TripID is not in the Trips table.
24. Count them and name the value.
25. State what it means, in business terms, that these tickets exist.

EVIDENCE

Count, value, and what it implies about the business process.

Count: 23

Value: PHP 3,222.24

These tickets appear in the record in TripID.

DECISION

Written, in full sentences.

The tickets should not be deleted. It might be an error on the system, but it is still part of a sale of a ticket which is important.

Applied steps renamed? NO Query name: Merge1

BAND C · TASK C3 · 12 POINTS

A diagnostic — find it, count it, say what it costs

The people nobody scheduled

26. Compare the Crew table against the DriverID and ConductorID columns in Trips.
27. Count the crew members who never appear on any trip.
28. State what those crew members have in common.

EVIDENCE

The count, and the pattern.

Count: 14

Almost all of them are conductors. There is only 1 driver who never appeared on any trip.

DECISION

Written, in full sentences.

The 14 crew members are listed in the Crew table but never appear as either a DriverID or ConductorID in the Trips table. This means they have no recorded trip assignments and should be reviewed to determine whether they are inactive, unassigned, or missing from the trip records.

Applied steps renamed? NO Query name: Append1

Name: _____ Drawn at: _____ Finished: _____

BAND D · TASK D1 · 15 POINTS

A judgement — decide, and defend it in writing

The trips that look wrong

29. Run an outlier test on FuelLitres in Trips.
30. Report the quartiles and the upper fence, and how many trips exceed it.
31. Decide: fix, exclude, escalate or accept. Defend it in writing.

EVIDENCE

Q1, Q3, the fence, the count beyond it, and your written decision.

DECISION

Written, in full sentences.

Applied steps renamed? YES / NO Query name: _____

Name: _____ Drawn at: _____ Finished: _____

BAND D · TASK D2 · 15 POINTS

A judgement — decide, and defend it in writing

Fact or dimension?

32. Open BayanihanBus_Maintenance.xlsx.
33. Decide whether it is a fact table or a dimension table.
34. If it is a fact, declare its grain as a sentence. If a dimension, name the fact it serves.
35. Defend the decision in writing.

EVIDENCE

Your classification, the grain sentence or join key, and the defence.

DECISION

Written, in full sentences.

Applied steps renamed? YES / NO Query name: _____

Name: _____ Drawn at: _____ Finished: _____

BAND D · TASK D3 · 15 POINTS

A judgement — decide, and defend it in writing

Should these two tables be joined?

- 36. Tickets carries TripID. Trips has one row per TripID.
- 37. Decide whether you would create a direct relationship from Tickets to Trips.
- 38. Defend the decision in writing, naming what you would do instead if you decline.

EVIDENCE

Your decision and its defence.

DECISION

Written, in full sentences.

Applied steps renamed? YES / NO Query name: _____